

- Overview
- Architecture & Tech Decisions
- System Architecture
- Dialog Management
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- Evaluation
- STT & Privacy
- Error Analysis
- Before / After & Outlook

PROBLEM DEFINITION

VocaDine Germany

VOICE RESTAURANT ASSISTANT FOR ENGLISH-SPEAKING TOURISTS

Sarah, 28, arrives in Haigerloch. She speaks no German. She says: *"Italian food for two, nothing too fancy"* — the assistant asks two follow-ups and reads out the top 3 restaurants. End-to-end: ~7 seconds.

600

ANNOTATED UTTERANCES

0.881

NER MACRO-F1

2058+

GERMAN CITIES

90%

TASK COMPLETION

SPACY NER

TF-IDF

VADER SENTIMENT

GOOGLE PLACES API

EDGE-TTS

VOCADINE VS. GENERIC ASSISTANTS — SCOPED COMPARISON

Alexa is stronger in ecosystem, hardware, multilingual support. This compares one scenario: English tourist, small German city.

FEATURE	ALEXA	VOCADINE
German city STT	Fails on small cities	2058+ + phonetic aliases
0-results handling	Empty list	Probe → slot rollback
Dialog flexibility	Skill-dependent	Implicit 5 slots/utterance
Ranking signal	Popularity only	TF-IDF + rating
Impatience / ambiguity	Not available	VADER + clarification

Ukpabi et al. (2019): voice assistants are an increasingly primary interface for local discovery in tourism.

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Architecture & Technology Decisions

END-TO-END PIPELINE WITH JUSTIFIED TECHNOLOGY CHOICES



COMPONENT	CHOICE	WHY
STT	SpeechRecognition · Google Web Speech	Free tier, reliable en-US, minimal setup
NLU	spaCy NER + rules	Trained model generalises; rules cover closed-domain edge cases
TTS	edge-tts · Neural	pytsx3 had Spyder/IPython env incompatibilities; edge-tts: neural quality, no API key
Ranking	scikit-learn TF-IDF + Rating	Deterministic, explainable, no extra API cost — limitations acknowledged
API	Google Places API	Live data, rich fields (reviews, attributes), free quota sufficient

NLU EXTRACTION CASCADE

- `extract_all_spacy(text)` — *NER model, all 5 labels*
- `GERMAN_CITIES` whitelist — *location fallback*
- `keyword maps` — *budget / occasion / datetime*

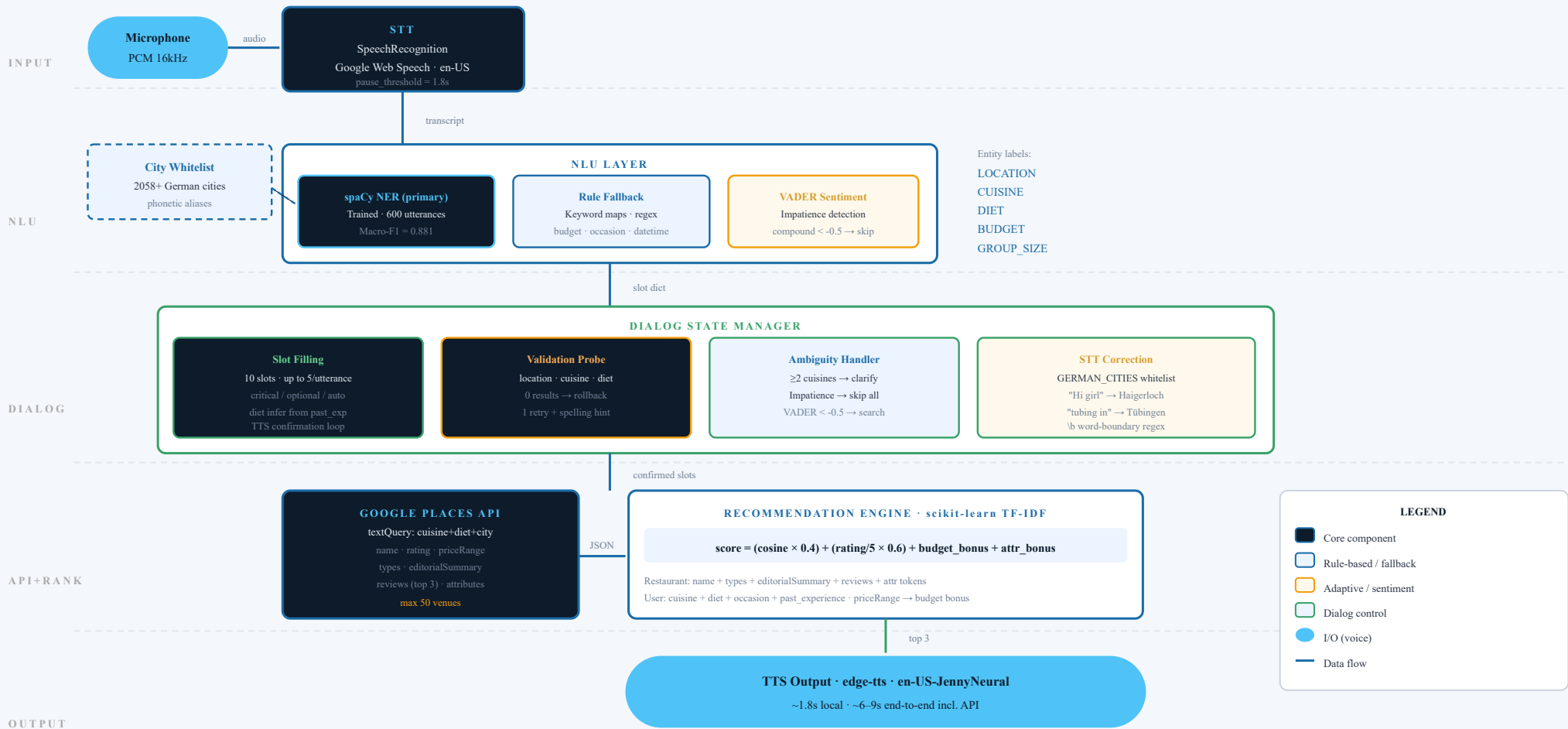
NLU is layered: the trained spaCy model runs first on every utterance. Only slots it misses are passed to the rule layer. This means unseen phrasings ("I fancy sushi", "plant-based") are handled without a keyword entry.

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DESIGN DOCUMENT

System Architecture

COMPONENT BOUNDARIES · DATA FLOWS · SUB-SYSTEM DETAIL



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IMPLEMENTATION

Dialog Management

IMPLICIT SLOT FILLING · VALIDATION · ADAPTIVE PACING

Input: "Italian food in Berlin for two tonight, nothing too fancy"
Extracted: location=Berlin · cuisine=Italian · group_size=2 · datetime=tonight · budget=moderate
Next question: "Any dietary restrictions?" (only remaining critical slot)

- Implicit filling:** up to 5 slots extracted per utterance — already-filled slots skipped automatically
- Validation probe:** after high-impact slot (location, cuisine, diet) → quick Places API call; 0 results → slot rolled back, re-prompted
- Impatience detection:** VADER compound < −0.5 OR "stop asking" → all remaining slots = "any", immediate search (Hutto & Gilbert, 2014)
- Ambiguity resolution:** ≥2 cuisines in one utterance → "I noticed Italian and Greek — which do you prefer?"
- Diet inference:** meat keyword in past_experience without negation → diet="none" inferred, question skipped

10 CONVERSATION SLOTS

SLOT	TYPE	EXAMPLE
location	CRITICAL	"I'm in Berlin"
cuisine	CRITICAL	"Italian"
diet	CRITICAL	"vegan" / "none"
budget	CRITICAL	"moderate"
past_experience	OPEN	"loved the pasta place"
datetime	OPTIONAL	"tonight"
group_size	OPTIONAL	"four"
occasion	OPTIONAL	"a date"
distance	OPTIONAL	"walking distance"
special_features	OPTIONAL	"outdoor seating"

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NLU: spaCy NER Training

600 ANNOTATED UTTERANCES · 5 ENTITY LABELS · BLANK ENGLISH MODEL

WHY A TRAINED MODEL?

Rule-based matching achieves perfect scores on utterances it was designed for — but fails silently on unseen phrasings. A trained NER model generalises: "I fancy sushi", "plant-based options", "we're counting pennies" are recognised without a keyword entry.

DATASET SECTION	COUNT
Single-label utterances	290
Multi-label combinations (2–4 slots)	270
Adversarial / edge cases	40
Total · split 70/15/15 (seed=42)	600

Annotation guidelines: data/annotation_guidelines.md — span rules, edge cases per label, ASR noise handling, inter-annotator agreement target $\kappa \geq 0.85$

TRAINING SETUP

- Model:** `spacy.blank("en")` — no pre-trained vectors, NER pipe only
- 40 epochs**, dropout=0.35, compounding batch 4→32
- Best checkpoint** saved at epoch 23 — val Macro-F1 = 0.9685
- Labels:** LOCATION · CUISINE · DIET · BUDGET · GROUP_SIZE

```
TRAIN_NLU.PY — TRAINING LOOP

for epoch in range(1, N_EPOCHS+1):
    random.shuffle(train_examples)
    for batch in minibatch(train_examples, ...):
        nlp.update(batch, drop=0.35, ...)
    if val_f1 >= best_f1: nlp.to_disk(MODEL_PATH)
```

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RECOMMENDATION MECHANISM

TF-IDF Recommendation Engine

SCIKIT-LEARN · FEATURE ENGINEERING · HONEST LIMITATIONS

SCORING FORMULA

```
score = (cosine_sim × 0.4) + (rating/5.0 × 0.6)
       + budget_bonus   (+0.1 if priceRange matches)
       + attribute_bonus (+0.1 outdoor / groups)
```

FEATURE ENGINEERING

- **Restaurant:** name + types + editorialSummary + top-3 review texts + attribute tokens ("outdoor seating terrace", "good for groups")
- **User:** cuisine + diet + occasion + past_experience free text
- **Why 40/60?** Design decision: Google's crowd-sourced rating correlates more reliably with overall quality than sparse keyword overlap on short descriptions
- **priceRange vs. price_level:** Google Places returns priceRange.endDate (actual € amount) — more precise than the 1–4 price_level scale. Used as a ±0.1 bonus signal rather than a TF-IDF feature because price is a hard constraint, not a preference to be weighted against text similarity

Known limitations: When Places API returns only generic types (e.g. "restaurant", "food"), TF-IDF degrades — reviews and editorial summary become the primary signal. TF-IDF cannot capture semantic similarity: "sushi" and "Japanese" are unrelated tokens unless co-occurring.

On the "training" claim: fit_transform() at runtime computes IDF weights from the live result set. This is lightweight on-demand vectoriser fitting — not equivalent to offline model training with a large corpus. The genuine ML training component is the **spaCy NER model** (Slide 5).

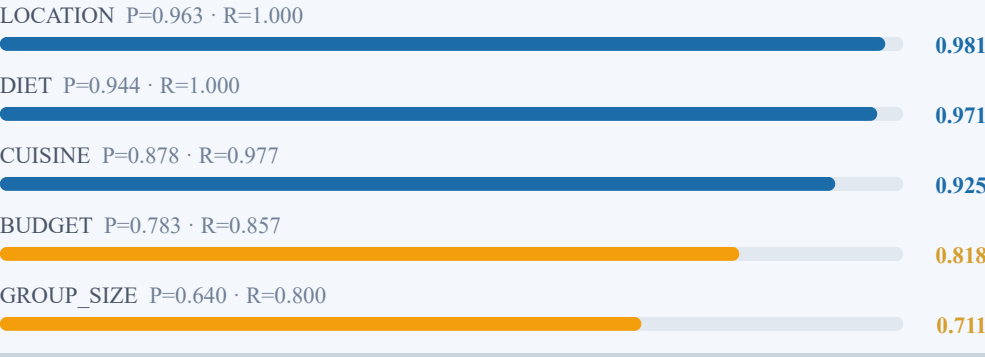
Rafailidis & Manolopoulos (2019): targeted ranking mechanisms produce significantly more relevant recommendations than popularity-only. Top 3 results: keeps response concise and actionable.

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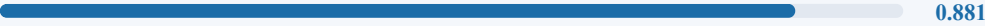
Evaluation

NER TEST SET (90 HELD-OUT UTTERANCES) · SYSTEM METRICS (END-TO-END)

NER — PER-LABEL F1



MACRO



GROUP_SIZE F1=0.711: implicit expressions ("me and my wife") are not annotatable NER spans — rule-based layer covers these. Architectural boundary, not model failure.

Val-F1 vs. Test-F1: Val-F1 (≈ 0.97 at best epoch) is the checkpoint-selection signal — the model is saved precisely when this peaks, making it optimistic by design. Test-F1 = 0.881 is evaluated on a fully held-out set the model never influenced. The gap is expected and methodologically sound.

SYSTEM METRICS

Metric	Value	Scope
Task Completion	90% (18/20)	Full conversation → top-3
Local latency	~1.8s	STT result → ranking → TTS start
End-to-end latency	~6–9s	Mic open → user hears result (incl. 2× Places API)
WER short words	~20%	"no", "two" — known STT limit

Latency note: The "~7s" scenario on Slide 1 is end-to-end including two Google Places API roundtrips. The 1.8s figure is local processing only. Both are real measurements — they measure different things.

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STT & Privacy

PHONETIC ALIAS SYSTEM · HONEST DATA FLOW ASSESSMENT

PHONETIC ALIAS SYSTEM (CORE DIFFERENTIATOR)

SPOKEN	STT OUTPUT	ALIAS FIX
Tübingen	"tubing in"	→ Tübingen
Haigerloch	"Hi girl"	→ Haigerloch
Horb am Neckar	"hope America"	→ Horb am Neckar
Döner	"doona"	→ Döner (cuisine)
Bayreuth	"Beirut"	→ Bayreuth

Word-boundary regex `\b . . \b` prevents substring false positives ("burg" inside "hamburg").

PRIVACY — HONEST DATA FLOW

DATA	DESTINATION	MITIGATION
Audio	Google Cloud (STT)	Mic only active during <code>listen()</code>
Location + preferences	Google Cloud (Places API)	Query string only, no user profile stored
Session slots	Local memory	Not written to disk, lost on exit
API key	.env file	Excluded from repo via .gitignore

"No always-on microphone" and "no persistent storage" are the minimum bar for a prototype — not differentiating privacy features. For production: GDPR Art. 13 disclosure required before first use, as audio and location data leave the device.

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ERROR ANALYSIS

Error Analysis

13 DOCUMENTED FAILURES · 10 RESOLVED · 3 OPEN OR KNOWN LIMITS

#	CATEGORY	EXAMPLE	ROOT CAUSE	STATUS
1	STT phonetic	"Tübingen" → "tubing in"	en-US unfamiliar with small city	✓ Alias
2	STT phonetic	"Döner" → "doona"	German food term	✓ Alias
3	Substring FP	"germany" → cuisine=German	Missing word boundary	✓ \b-regex
4	Negation missed	"no restrictions" → diet=None	Positive keywords only	✓ Pattern added
5	Impatience ignored	"stop asking" → continued	No sentiment layer	✓ VADER added
6	Ambiguity silent	"Italian or Greek?" → Italian	First-match-wins	✓ Clarification
7	Group size wrong	"me and my family" → 2	Pattern too generic	✓ family=4
8	Budget keyword gap	"average price" → unrecognised	Missing in map	✓ Extended
9	Diet inference FP	"loved the meatballs" → diet=none	Meat without negation	✓ Inference logic
10	0-results silent	Unknown city → continues	No retry	✓ Probe + retry
11	Diet inference limit	"served meat and I don't like meat"	Clause-separated negation	⚠ Known limit
12	Venue type wrong	Shisha bar in results	No type filtering	✗ Phase 3
13	Geographic imprecision	Berlin-Zehlendorf → Moabit	City only, no radius	✗ Phase 3

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Before / After & Phase 3 Outlook

WHAT CHANGED BETWEEN PHASE 1 AND PHASE 2

DIMENSION	PHASE 1	PHASE 2
NLU	Rule-based only	spaCy NER + rule fallback
NER Macro-F1	—	0.881 (90 held-out)
Training data	None	600 utterances, 420/90/90
TTS	pyttsx3 (env issues)	edge-tts · Neural voice
Slot filling	Sequential, 1/utterance	Implicit, up to 5/utterance
Impatience	Ignored	VADER detection
Errors documented	~5	13 · 10 resolved
Task completion	Not measured	90% (18/20)

PHASE 3 ROADMAP

- Type filtering** — exclude non-restaurant results (shisha bars, clubs) from Places API
- Radius search** — configurable km radius around city centre
- Multilingual** — German, French, Spanish speaker profiles
- Offline fallback** — local model when no internet available

What remains limited: TF-IDF on sparse type strings degrades without reviews; GROUP_SIZE and BUDGET NER still benefit from rule augmentation; end-to-end latency (~7s) acceptable for prototype, not production-ready; privacy disclosure would need GDPR basis for real deployment.